

HW4: IP

Part 1 Basic IPv4

1. It is the very first packet sent by the traceroute, which always starts at 1.
2. 20 bytes. It says "Header Length: 20 bytes" right in the packet details.
3. 36 bytes. Total length is 56 bytes minus the 20-byte IP header = 36.
4. No. The Flags field is 0x0000 and the Fragment offset is 0.
5. Identification (needs a new ID for each packet), TTL (traceroute increments this by 1 each time), and Header Checksum (math has to be recalculated because ID and TTL change).
6. Version, Header Length, Source IP, Dest IP, and Protocol. It is all the same session from your machine to the exact same server using UDP.
7. No. Different routers use different default starting TTLs (like 64 or 255) and they are different distances away, so the number left over when the ICMP error gets back to you will vary.

Part 2 IPv6

8. 2601:193:8302:4620:215c:f5aa:8040:27a
9. 2001:558:feed::1
10. 0x4d183
11. 39 bytes.
12. UDP.
13. 2
14. 2607:f8b0:4006:806::200e